// EFFAT UNIVERSITY · COLLEGE OF ENGINEERING

CS3081 · ARTIFICIAL INTELLIGENCE

RESOLUTION HACKATHON

> case_file: the_missing_exam_paper

In-class team sprint · 50 min · 4 levels · 100 XP + bonus

Mentor: Dr. Naila Marir

incident_report.txt

```
$ cat incident_report.txt

Monday 08:00. The CS3081 final exam paper is gone from the instructor's office.
Three students were seen near the office on Sunday night:
  → Sara   → Omar   → Lina

Campus security left 7 clues. Your team has 50 minutes to hack the case with
resolution:
find who took it, who is innocent, and which accusation cannot be proven.

$ ./start_hackathon --teams 3-4 --mode proof_by_contradiction
```

// TEAM REGISTRATION

TEAM NAME	Tigers	TABLE #	
HACKER 1	Osama	HACKER 2	Mohammed
HACKER 3	Rawad	HACKER 4	Muzzamil

// THE 4 LEVELS

LEVEL 01
CNF Relay
10 min · 20 XP

LEVEL 02
Resolution Dominoes
10 min · 30 XP

LEVEL 03
Solve the Case
20 min · 40 XP

LEVEL 04
Final Accusation
10 min · 10 XP

Bonus XP: +3 FIRST BLOOD per level · +3 BUG HUNTER per accepted objection · +3 SIDE QUEST in Level 03

README.md

Rules of the Hack

50 MIN | 100 XP

▶ Pick your roles — rotate every level

C
CONVERTER
Turns logic into CNF: eliminate ↔ and →, De Morgan, distribution.

R
RESOLVER
Hunts complementary literals and builds new clauses (resolvents).

✓
CHECKER
Debugs every step and writes the final answer. Nothing ships without the Checker.

▶ How the game runs

01 Teams of 3–4. Play the four levels in order.

02 When your team clears a level, raise your hand. The mentor checks it and unlocks the next level.

03 **First Blood:** the first team to clear each level correctly gets **+3 XP**.

04 Allowed moves only: eliminate ↔, eliminate →, De Morgan, distribution, resolution and factoring.

05 Show every step. A correct answer with no steps earns only half the XP.

▶ XP table

LEVEL	CHALLENGE	TIME	XP
01	CNF Relay — convert the clues	10 min	20
02	Resolution Dominoes — match clause cards	10 min	30
03	Solve the Case — proof by contradiction	20 min	40
04	Final Accusation — present and object	10 min	10 + steals
	Total	**50 min**	**100 + bonus**

DATASET

The Evidence Board

10 SYMBOLS | 7 CLUES

▶ Symbol table

SYMBOL	MEANING
D	The office door was unlocked
S	Sara took the exam paper
O	Omar took the exam paper
L	Lina took the exam paper
C	The security camera was switched off
F	The office door was forced open
Y	Lina had a copy of the office key
K	Khalid lent his office key
A	The alarm rang

▶ Clues from campus security

#	CLUE (ENGLISH)	LOGIC
1	If the door was unlocked, then Sara or Omar took the paper.	D → (S ∨ O)
2	The door was unlocked.	D
3	If Sara took the paper or the door was forced, then the camera was off.	(S ∨ F) → C
4	The camera was NOT off (it recorded all night).	¬C
5	Lina could only have taken it if she had a copy of the key.	L → Y
6	Lina had a copy of the key if and only if Khalid lent his key.	Y ↔ K
7	It is not true that Khalid lent his key AND the alarm rang.	¬(K ∧ A)

CHEAT CODE

To show KB ⊨ α: add **¬α** to the KB, convert everything to CNF, and keep resolving. If you produce the empty clause **( )**, it's a contradiction, so KB ⊨ α. If no new clauses can be made and there is no ( ), there is **no entailment**.

LEVEL 01 / 04

CNF Relay

10 MIN 20 XP

RELAY Pass the sheet around the team. **Each hacker writes ONE step, then passes it on.** Write the rule you used next to each step.

▶ Clue 1: D → (S ∨ O) [4 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	¬D ∨ (S ∨ O)	Eliminate →	Mohammed
2	¬D ∨ S ∨ O	Associativity	Rawad
3	(¬D ∨ S ∨ O) → 1 clause	Final CNF	Muzzamil

▶ Clue 3: (S ∨ F) → C [6 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	¬(S ∨ F) ∨ C	Eliminate →	Osama
2	(¬S ∧ ¬F) ∨ C	De Morgan	Muzzamil
3	(¬S ∨ C) ∧ (¬F ∨ C)	Distribution	Mohammed

▶ Clue 6: Y ↔ K [6 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	(Y → K) ∧ (K → Y)	Eliminate ↔	Rawad
2	(¬Y ∨ K) ∧ (¬K ∨ Y)	Eliminate →	Osama
3	2 clauses: (¬Y ∨ K), (¬K ∨ Y)	Final CNF	Muzzamil

▶ Clue 7: ¬(K ∧ A) [4 XP]

STEP	EXPRESSION	RULE USED	WRITTEN BY
1	¬K ∨ ¬A	De Morgan	Muzzamil
2	(¬K ∨ ¬A) → 1 clause	Final CNF	Mohammed
3	—		

▶ Compile the final KB in CNF

List every clause from all seven clues, one per row, as K1, K2, K3, … You need these numbers in Level 03.

NO.	CLAUSE	NO.	CLAUSE
K1	¬D ∨ S ∨ O	K6	¬L ∨ Y
K2	D	K7	¬Y ∨ K
K3	¬S ∨ C	K8	¬K ∨ Y
K4	¬F ∨ C	K9	¬K ∨ ¬A
K5	¬C	K10	(reserved: ¬goal in L03)

LEVEL 02 / 04

Resolution Dominoes

10 MIN | 30 XP

Cut out the 12 clause cards on the next page and shuffle them. Then:

01 Find pairs of cards that share a **complementary literal** (like P and ¬P). Each card belongs to exactly one pair.

02 Place the two cards side by side like dominoes and write the resolvent below.

03 If the resolvent has a duplicate literal, apply **factoring** and say so.

⚠ BUG

One pair is a **trap**. It looks like it gives ( ) but it doesn't. Think carefully about what one resolution step is allowed to do.

PAIR	CARD + CARD	RESOLVED ON	RESOLVENT	FACTORING?
1	B + G	T	U ∨ U ⇒ U	(Y) / N
2	C + H	Q	P ∨ R	Y / (N)
3	D + J	Y	X ∨ Z ∨ W	Y / (N)
4	E + L	M	( ) empty clause	Y / (N)
5	F + K	A	B ∨ S ∨ R ∨ S ⇒ B ∨ R ∨ S	(Y) / N
6	A + I (TRAP)	G	H ∨ ¬H (tautology, NOT ( ))	Y / (N)

▶ Bug report: the trap pair

Which pair is it? Explain in one or two sentences why it does NOT give the empty clause.

Pair 6: CARD_A (G ∨ H) + CARD_I (¬G ∨ ¬H). One resolution step may cancel only ONE complementary pair. Resolving on G gives (H ∨ ¬H), a tautology (always true), not ( ). Cancelling G and H in the same step is an illegal move, so this pair does not give ( ).

LEVEL 02 · ASSETS

Clause Cards

CUT ON DASHED LINES

CARD_A

(G ∨ H)

{ resolve me }

CARD_B

(¬T ∨ U)

{ resolve me }

CARD_C

(¬Q ∨ R)

{ resolve me }

CARD_D

(X ∨ ¬Y ∨ Z)

{ resolve me }

CARD_E

(M)

{ resolve me }

CARD_F

(A ∨ B ∨ S)

{ resolve me }

CARD_G

(T ∨ U)

{ resolve me }

CARD_H

(P ∨ Q)

{ resolve me }

CARD_I

(¬G ∨ ¬H)

{ resolve me }

CARD_J

(Y ∨ W)

{ resolve me }

CARD_K

(¬A ∨ R ∨ S)

{ resolve me }

CARD_L

(¬M)

{ resolve me }

LEVEL 03 / 04

Solve the Case

20 MIN | 40 XP

Use your CNF KB (K1, K2, ...) from Level 01. For each case, add the **negation** of what you want to prove, then resolve. Number new clauses R1, R2, R3, ...

▶ Case 3A — Did Omar take the paper? Prove KB ⊨ O [15 XP]

ADD TO KB: **K10: ¬O**

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	K1	K2	D	S v O
R2	R1	K10	O	S
R3	K3	K5	C	¬S
R4	R2	R3	S	**( ) empty clause**
R5	—	—	—	—

VERDICT: **( ) derived ⇒ KB ⊨ O. Omar took the paper.**

▶ Case 3B — Is Sara innocent? Prove KB ⊨ ¬S [10 XP]

ADD TO KB: **K10: S**

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	K10	K3	S	C
R2	R1	K5	C	**( ) empty clause**
R3	—	—	—	—

VERDICT: **( ) derived ⇒ KB ⊨ ¬S. Sara is innocent.**

LEVEL 03 / 04 · BOSS FIGHT

Case 3C — What about Lina?

15 XP

interrogation.log

```
Omar: "Lina did it with me!"
Lina: "I am innocent!"
$ try prove(L) && try prove(¬L)
```

▶ Attempt 1 — add ¬L and resolve

K10: ¬L

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	K6	K7	Y	¬L ∨ K
R2	R1	K9	K	¬L ∨ ¬A
R3	K4	K5	C	¬F
R4	—	—	—	Saturated: no new clauses, no ( )

▶ Attempt 2 — add L and resolve

K10: L

STEP	CLAUSE 1 (NO.)	CLAUSE 2 (NO.)	RESOLVED ON	NEW CLAUSE (RESOLVENT)
R1	K10	K6	L	Y
R2	R1	K7	Y	K
R3	R2	K9	K	¬A
R4	—	—	—	Stuck: ¬A has no partner, no ( )

▶ Conclusion

Can the detectives accuse Lina? Can they clear her? What does this mean about entailment?
Hint: look at where the literal L appears in the KB.

No to both. L appears only once in the KB, as ¬L in K6 (¬L ∨ Y), so ¬L can never be cancelled.

Neither attempt reaches ( ): KB ⊭ L and KB ⊭ ¬L. Lina can be neither accused nor cleared.

L is independent of the KB (true in some models, false in others), so Omar's claim is unproven.

SIDE QUEST Finished early? Prove that the door was **not** forced: KB ⊨ ¬F. Write your proof on the back of this page for **+3 XP**.

★ SIDE QUEST — Prove KB ⊨ ¬F. Add K10: F

R1: K10 + K4 on F → C

R2: R1 + K5 on C → ( )

( ) derived ⇒ KB ⊨ ¬F. The door was not forced.

LEVEL 04 / 04

Final Accusation

10 MIN | 10 XP

01 The mentor picks one case (3A, 3B or 3C) for your team. Present it on the board in **2 minutes**.

02 While a team presents, the other teams are **code reviewers**. They look for an invalid step.

03 To object, hold up your **OBJECTION!** card and name the exact step and the broken rule.

04 A **correct objection steals 3 XP** from the presenting team. A wrong objection costs 2 XP.

05 A clean presentation with no accepted objection earns the full 10 XP.

▶ Code review log

AGAINST TEAM	STEP	WHAT WAS WRONG	ACCEPTED?

▶ Post-mortem (individual, one sentence each)

Why is resolution called a “proof by contradiction”? What does it mean when no empty clause can ever be produced?

It assumes the opposite of the goal (adds ¬α) and shows KB ∧ ¬α derives ( ), a contradiction, so α must hold.

If no empty clause can ever be produced (no new clauses remain), KB ∧ ¬α is satisfiable,

so KB does not entail α: the goal cannot be proven from the KB.

UNLOCKABLES

Achievements & Leaderboard

TICK WHAT YOU EARN



FIRST BLOOD

First team to clear a level. +3 XP each time.

□ earned



TRAP DETECTOR

Spotted and explained the trap pair in Level 02.

□ earned



SIDE QUEST

Proved KB ⊨ ¬F in Level 03. +3 XP.

□ earned



CLEAN BUILD

Presented in Level 04 with zero accepted objections.

□ earned

▶ Team scoreboard (mentor use)

L01	L02	L03	L04	BONUS	STEALS	TOTAL
/20	/30	/40	/10	+	±	**/100**

// cut out and hold up during Level 04

ERROR 403: INVALID STEP

OBJECTION!

step ____ breaks rule: _____